Ayan Khan

AyanKhanOffiiciall@gmail.com
Portfolio | GitHub | LinkedIn | LeetCode

Profile

Bachelor of Computer Applications student specializing in Machine Learning and Data Science, with a strong foundation in software development, object-oriented programming, algorithms, and computational problem solving. Experienced in building modular C++ applications involving stochastic modeling, image processing, and compiler design. Interested in developing reliable software, exploring computer science concepts, and contributing to real-world engineering projects.

Education

Bachelor of Computer Applications – Machine Learning and Data Science 2025 – 2028
Deen Dayal Upadhyaya Gorakhpur University, India SGPA: 9.08/10.00
Relevant Study: Data Structures and Algorithms, Object-Oriented Programming, Statistics, Exploratory Data Analysis, Databases, Machine Learning Fundamentals

Technical Skills

Languages: C++, Python, C, Java; JavaScript fundamentals
Core: Data Structures and Algorithms, Object-Oriented Programming, Complexity Analysis, Modular Design, Debugging, Testing, Error Handling
Scientific and Data: Markov Chains, Stochastic Modeling, Image Processing Fundamentals, Pandas, NumPy, Matplotlib, SQL, Exploratory Data Analysis
Tools: Linux, Git, GitHub, VS Code, Command Line, Jupyter Notebook, Google Colab, C++ STL

Projects

Markov Text Generator *C++, Markov Chains, CLI*

- Built a modular text generator using fixed-order Markov chains, probabilistic token sampling, and state-transition modeling.
- Implemented tokenization, model training, transition-table construction, and generation as independently testable components.

Asciify-cpp *C++, Image Processing, CLI*

- Developed a command-line tool that converts images into grayscale and color ASCII-art representations.
- Implemented configurable output dimensions and pixel-intensity mapping with an emphasis on portability and straightforward usage.

Prometheus Compiler *C++, Compiler Design*

- Developing an experimental compiler with lexical analysis, tokenization, parsing, and Abstract Syntax Tree generation.
- Applied modular architecture, object-oriented programming, error handling, debugging, testing, and Git-based version control.

Technical Activities and Achievements

- Solved 223+ LeetCode problems and achieved a 1500+ contest rating, strengthening algorithm design, complexity analysis, and debugging skills.
- Maintained a repository of 200+ C++ and Python solutions with documented approaches and time-space complexity analysis.
- Completed the First Contributions GitHub workflow using forks, branches, commits, pull requests, and code review.